

Task 1

$$N = 512$$

S = sparsity factor (zeros)

$$[N \times N] \times [N \times 1] = [N \times 1]$$

(a) MVM dense.

For an output with N elements, we need to perform $N \times N \times 1$ MACs, that comes out to be $2N^2$ FLOPs since one MAC is 2 FLOPs.

$$\text{dense MVM FLOPs} = 2N^2$$

(b) dense MVM bytes.

for an output size $N \times 1$,

we need N elements from matrix A and the weight matrix with N elements.

So we will need N^2 elements that gives us $4N^2$ bytes as each element is FP32 which is 4 bytes each.

$$\text{dense MVM bytes} = 4N^2$$

(c) Sparse compute

With sparsity s , the no of non zero elements $2(1-s)N^2$

with sparsity s (fraction of zeroes) the non-zero fraction is $(1-s)$

$$nnz = (1-s) \cdot N^2$$

MAC count

with perfect sparsity exploitation

$$MACs = nnz = (1-s) \cdot N^2$$

Each MAC is 2 FLOPs

FLOPs

$$F_{sparse} = 2 \cdot (1-s) \cdot N^2$$

(d) Sparse Memory

In CSR we need 3 Arrays

The non zero weights

One F32 per non zero element

$$\text{length} = (1-s)N^2$$

$$\text{Bytes} = 4(1-s)N^2$$

column position of each non-zero.

One INT 32 per non zero

$$\text{length} = (1-s)N^2$$

$$\text{Bytes} = 4(1-s)N^2$$

Row pointer.

One INT32 per row, plus one
terminating length = $N+1$

Independent of Sparsity length = $N+1$

$$\text{Bytes} = 4(N+1)$$

$$M_{\text{space}} = 4(1-s)N^2 + 4(1-s)N^2 + 4(N+1)$$

$$= 8(1-s)N^2 + 4(N+1)$$

$$\text{Sparse MVM bytes} = 8(1-s)N^2 + 4(N+1)$$

Task 2

FLOPs speedup expression
and s for 2X Speedup.

Speedup is defined as the
ratio of dense work to Sparse work.

$$\text{Speedup} = \frac{F_{\text{dense}}}{F_{\text{sparse}}}$$

$$= \frac{2N^2}{2(1-s)N^2}$$

$$\boxed{\text{Speedup} = \frac{1}{1-s}}$$

For 2X Speedup.

$$2 = \frac{1}{1-s}$$

$$2(1-s) = 1$$

$$1-s = \frac{1}{2}$$

$$\boxed{s = 0.5}$$

At 50% sparsity, sparse MVM is 2x faster
than dense mvm.

Task 3 Memory breakeven sparsity.

To find the breakeven point, we set both dense MVM and sparse MVM bytes as equal

$$8N^2(1-s) + 4(N+1) = 4N^2$$

divide by $4N^2$

$$2(1-s) + \frac{4(N+1)}{4N^2} = 1$$

$$2s = (2-1) + \frac{N+1}{N^2}$$

$$2s = 1 + \frac{N+1}{N^2}$$

divide by 2 to get final expression

$$s = \frac{1}{2} + \frac{N+1}{2N^2}$$

above this 's'
sparse MVM
uses less
memory.

for the given case with $N = 512$

$$s = 0.5 + \frac{513}{524,288}$$

$$s = 0.500979$$

Task 4 Hardware perfectly exploits sparsity.

End to end speedup at $s = 0.9$

Memory-bandwidth limited system with 320GB/s. $N = 512$

$$\text{Speedup} = \frac{T_{\text{dense}}}{T_{\text{sparse}}}$$

$$= \frac{M_{\text{dense}} / \text{BW}}{M_{\text{sparse}} / \text{BW}}$$

$$= \frac{M_{\text{dense}}}{M_{\text{sparse}}}$$

Dense

$$M_{\text{dense}} = 4N^2 = 1048576 \text{ B}$$

Sparse

$$M_{\text{sparse}}(0.9) = 8(1-s)N^2 + 4(N+1)$$

$$= 8 \times 0.1 \times 262,144 + 4 \times 513$$

$$= 211,767 \text{ B}$$

$$\text{Speedup} = \frac{1048576}{211767}$$

$$\boxed{\text{Speedup} = 4.95}$$

Even though the possible
Speedup through compute is

$$S_{\text{perf}} = \frac{1}{1-s} = \frac{1}{1-0.9} \\ = \frac{1}{0.1} = 10$$

We can only do 4.95 due to the
memory bandwidth limit.